CAPSTONE PROJECT REVIEW PROGRESS RECORD

Review:	**Review–1**	**Date of Review:**	**7 August 2026**
Project Title:	**Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators**		
Cohort:	_____	**Project Guide:**	Dr. Arun Nimmala, Associate Professor, Dept. of ECE

Student Details:

S. No.	Student Name	Student ID	Signature
1	Velivala Keerthi	2320040083	
2	T. Venkata Ram Charan Reddy	2320040017	
3	Shaik Hanifa	2320040009	
4	K. Dinesh	2320040095	

Brief Description of the Project

Briefly explain the premise of the project, the problem being addressed and the proposed approach. (less than 200 words)

Convolutional Neural Network (CNN) edge accelerators are constrained by the high area and power dissipation of Multiply-Accumulate (MAC) datapaths, where 8x8 exact multipliers dominate silicon cost (68 LUTs on Xilinx Zynq-7020). While approximate computing trades controlled arithmetic error for hardware savings, uniform approximation causes severe accuracy degradation or negligible gains. Existing layer-wise mapping methods rely on computationally prohibitive evolutionary search (ALWANN: 7.5 CPU-days) or GPU retraining loops (TRAM). This project proposes a lightweight, single-pass sensitivity metric computed in 0.2 seconds combining static weight variance and dynamic loss gradients via a 256-sample calibration probe without retraining. Layers are ranked such that error-vulnerable early layers receive exact multipliers while noise-resilient deep layers receive aggressive approximate multipliers. The heterogeneous design is mapped onto synthesizable Verilog Processing Elements (PEs) and deployed on a PYNQ-Z2 FPGA board via AXI4-Stream DMA, achieving up to 32.5% multiplier area reduction while preserving baseline classification accuracy.

Progress Achieved up to This Review

Briefly mention the work completed, implementation/simulation/experimental progress, results obtained and other major milestones achieved up to this review.

- **RTL Multiplier Library Design:** Designed four parameterizable 8x8 unsigned multiplier architectures in synthesizable Verilog: exact (full partial-product array), truncated (LSB column drop), compensated (constant midpoint rounding bias), and OR-based logic.
- **Exhaustive 65,536-Pair Simulation:** Verified all 256x256 input combinations in Icarus Verilog; extracted golden arithmetic error distributions (Mean Error, NMED, MRED, Max Absolute Error).
- **Vivado Out-of-Context Synthesis on Zynq-7020:** Obtained post-synthesis FPGA utilization targeting xc7z020clg400-1. Exact: 68 LUTs; Trunc-4: 56 LUTs (-17.6%); Comp-4: 56 LUTs (-17.6%); OR-4: 64 LUTs (-5.9%).
- **Circuit-Level Dominance Discovery:** Proved that Compensated strictly dominates Truncated at identical LUT cost (56 LUTs) with 35% lower error, as the synthesis tool absorbs the +2^(T-1) constant into existing carry logic at zero extra hardware overhead. OR-based design proved inferior and was discarded.
- **Literature Benchmark & Experimental Protocol:** Formulated formal evaluation protocol against ALWANN, TRAM, Sayadi et al., and Kumari & Rao, establishing strict Spearman gate criteria (|rho| >= 0.70) before committing to physical hardware RTL design.

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Review:	**Review–2**	**Date of Review:**	**3 October 2026**
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Cohort:	_____	**Project Guide:**	Dr. Arun Nimmala, Associate Professor, Dept. of ECE

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Progress Achieved up to This Review

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- **Baseline CNN & Affine INT8 Quantization:** Trained 4-conv baseline CNN on Fashion-MNIST (84.69% exact baseline accuracy). Developed post-training INT8 quantization engine (unsigned 8-bit activations [0, 255], sign-magnitude weights) matching unsigned hardware multipliers.
- **Single-Pass Sensitivity Formulation (S1–S5):** Formulated candidate sensitivity metrics. Extracted first-order loss gradients via TensorFlow GradientTape on a 256-sample calibration batch in 0.2s. Proved conv1 is critical (S=1.000) while conv4 is highly robust (S=0.014).
- **TRUNC=4 Sweep & Noise Floor Discovery:** Executed 12-cell validation sweep across 10,000 test images. Measured accuracy drops fell within ±0.1% to 0.3%, identifying the CNN resilience boundary and revealing an initial statistical noise floor (formal NO-GO at TRUNC=4).
- **TRUNC=6 Scaling & Signal Amplification:** Scaled truncation depth to T=6 (dropping 21 of 64 partial products, amplifying NMED by 6.6x to 0.001234, saving 32.5% area). Proof-of-concept tests demonstrated sensitivity assignment achieved 85.27% accuracy, recovering 81.5% of accuracy lost by uniform approximation.
- **Spearman Correlation & Formal Gate Clearance:** Proposed combined metric S3 achieved Spearman |rho| = 0.80 on TRUNC=6, clearing the pre-defined |rho| >= 0.70 GO gate and validating the core research hypothesis.

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Review:	**Review–3**	**Date of Review:**	**6 November 2026**
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Cohort:	_____	**Project Guide:**	Dr. Arun Nimmala, Associate Professor, Dept. of ECE

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Brief Description of the Project

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Progress Achieved up to This Review

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- **NSGA-II Multi-Objective Optimization Benchmark:** Implemented evolutionary search across all 256 configurations optimizing Accuracy Loss vs. LUT Cost. Confirmed our sensitivity-guided assignment lands at the Pareto knee in 0.2s, achieving a >3,000,000x speedup over genetic search (ALWANN: 7.5 CPU-days).
- **Complete Synthesizable Accelerator RTL:** Designed full Verilog datapath (axm_8x8.v, mac_unit.v, conv_pe.v, layer_controller.v, cnn_accelerator_top.v) with 2-bit runtime mode control and AXI4-Stream interfaces. Verified across 262,144 exhaustive test vectors with 100% pass (0 mismatches).
- **Automated Vivado Block Design:** Developed vivado_block_design.tcl automating Zynq-7020 PS/PL integration, AXI SmartConnect, and AXI DMA engine (MM2S & S2MM channels) clocked at 100 MHz.
- **PYNQ-Z2 On-Board Deployment Pipeline:** Extracted INT8 quantized weights (weights_int8.npy), built Python on-board driver (pynq_inference.py), and authored an interactive 7-cell Jupyter notebook (pynq_cnn_demo.ipynb) for hardware telemetry.
- **Publication Readiness & Code Release:** Generated all 7 IEEE publication figures, drafted 6-page IEEE manuscript, compiled comprehensive 20-slide presentation deck with embedded speaker notes, and published repository to GitHub.

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